

EW – 1 Lab 7

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ANDROID-ARDUINO INTEGRATION

Arduino being a versatile platform can be integrated with a lot of other projects. MIT AppInventor (MIT AI2) is a blocks-based android app development platform that can be used to develop basic android applications.

Activity – 1

AIM: To control LED on Arduino with an android application.

Components Required:

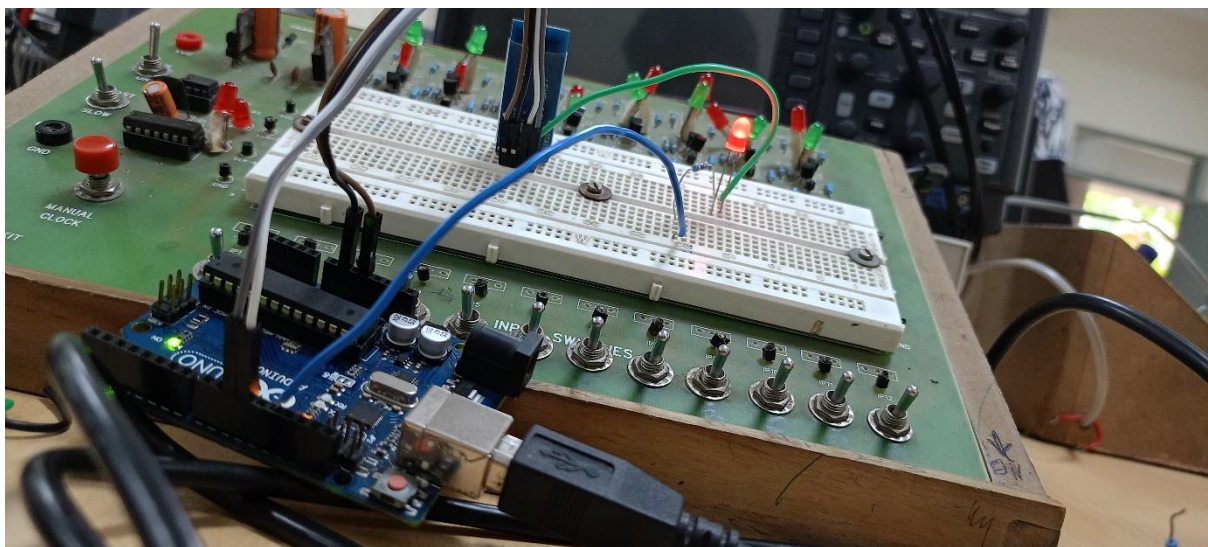
- Arduino Uno
- Breadboard
- LED
- 220 Ω resistor
- HC-05 Bluetooth Module
- Jumper wires
- An android smartphone (5.0+) and a laptop

Procedure:

- Plug the LED into the breadboard and connect its cathodes to GND of Arduino via the resistor.
- Connect digital pin 13 to the anode of the LED.
- Upload the code for activity 1.
- Import the project for activity 1 to MIT App Inventor and connect the companion app.
- Observe the output when the button is pressed in the app.

Results:

The LED turns on and off as the respective buttons are pressed.



Observation:

The signal generates from the smartphone Bluetooth module, is received by the HC-05 module, and then by Arduino, which responds by toggling the state of its LED pin.

Conclusion: LED control was successfully established using wireless Bluetooth connection between Arduino and Android.

Activity – 2

AIM: Display sensor readings from Arduino to Android app transmitted via Bluetooth module.

Components Required:

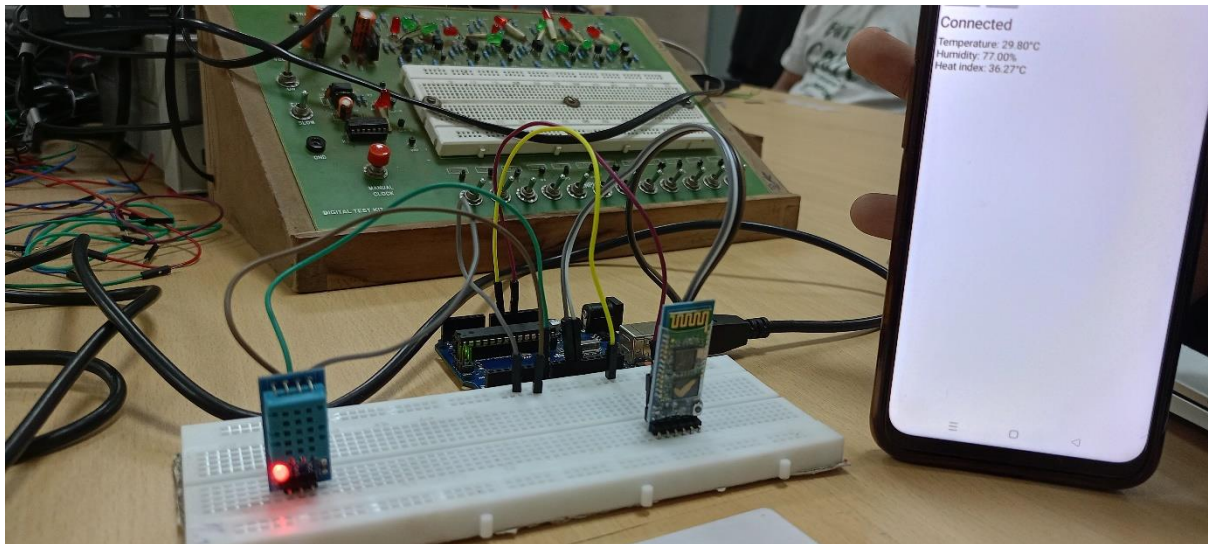
- Arduino Uno
- Breadboard
- DHT11 sensor module
- Jumper wires

Procedure:

- Plug the sensor into the breadboard and power it.
- Connect digital pin 7 to the data pin of the sensor.
- Upload the code for activity 2.
- Import the same project file to MIT App Inventor and connect the companion app.
- Observe the sensor output in the application.

Results:

The temperature and humidity data are displayed on the screen.



Observation:

The same wireless connection is established, but this time the transmission is the other way around.

Conclusion: The sensor data was successfully displayed on the android application screen.